GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2026.

GATE 2026

GATE 2026 | Q.1 | 1 mark

Among the following options, the antonym of the word 'nocturnal' is _____.

(A) normal
(B) diurnal
(C) abnormal
(D) exceptional

Answer: (B) [self-solved, no official key available]

Solution. Step 1. *Nocturnal* means active or occurring at night.
Step 2. The opposite, active during the day, is *diurnal*.

GATE 2026 | Q.2 | 1 mark

The statements (S1), (S2), and (S3) pertain to the scores obtained by students in an exam. The maximum possible marks in the exam is 150. (S1) The highest score is 100. (S2) The fourth highest score is 76. (S3) There are at least four students whose scores are within 25 of each other. Which one of the following options is necessarily correct?

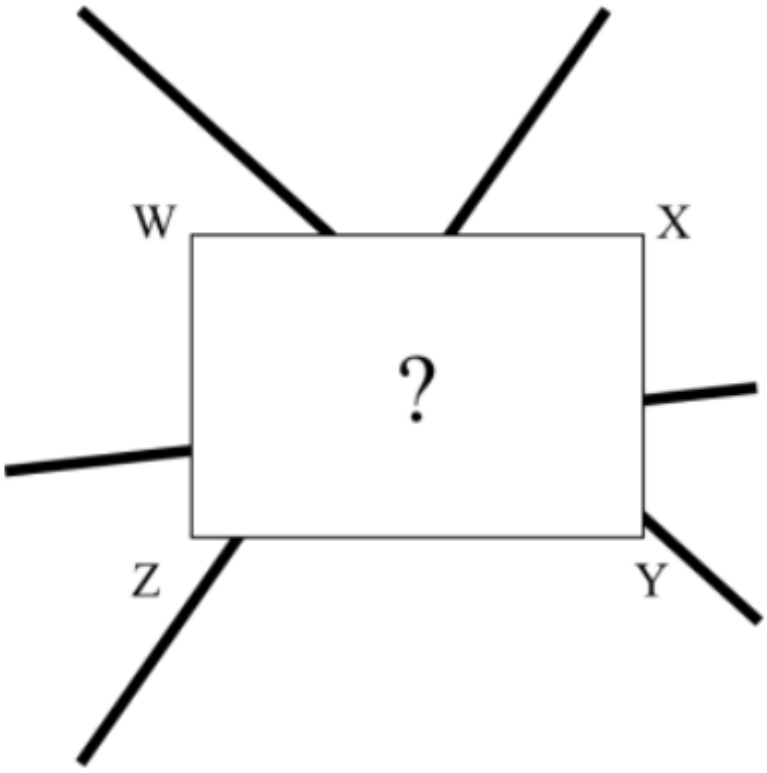
(A) (S1) and (S2) together imply (S3)
(B) (S1) and (S3) together imply (S2)
(C) (S2) and (S3) together imply (S1)
(D) (S1) implies (S3)

Answer: (A) [self-solved, no official key available]

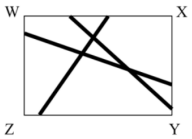
Solution. Step 1. If the highest score is 100 and the fourth highest is 76, the four highest scores all lie in $[76, 100]$.
Step 2. Any two of them differ by at most $24 < 25$, so at least four students have scores within 25 of each other: (S3) follows.

GATE 2026 | Q.3 | 1 mark

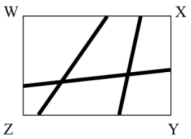
The figure below has exactly three intersecting line segments with a rectangular portion $WXYZ$ missing. Which one of the following options P, Q, R, and S is the missing portion $WXYZ$?



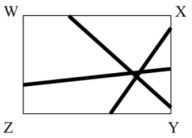
(A)



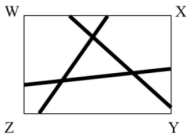
(B)



(C)



(D)



Answer: (C) **[UNVERIFIED: figure hard to read, check against official key]**

Solution. Step 1. The three segments must continue as straight lines through the missing rectangle $WXYZ$.
Step 2. Check each candidate: only option R makes all three segments enter and leave the rectangle collinearly. *(Low confidence.)*

GATE 2026 | Q.4 | 1 mark

Real numbers y, p, and n (all greater than 1) satisfy $\left(\log_{p^{1/n}} y\right)\left(\log_{y^{1/n}} p\right) = 16$, where the logarithms are taken to the bases $p^{1/n}$ and $y^{1/n}$. The value of n is _____.

(A) 2
(B) 4
(C) 8
(D) 16

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $\log_{p^{1/n}} y = n \log_p y$ and $\log_{y^{1/n}} p = n \log_y p$.
Step 2. Their product is $n^2 (\log_p y)(\log_y p) = n^2$.
Step 3. $n^2 = 16 \Rightarrow n = 4$.

GATE 2026 | Q.5 | 1 mark

The following observation is made about the scores obtained by 100 students in an exam: 'For each student, there exists another student in the class such that their scores are at most ten marks away.' If the above statement is false, which one of the following statements is necessarily true?

(A) For each student, the scores of all the other students are more than 10 marks away.
(B) There exists at least one student in the class for whom the scores of all the other students are more than 10 marks away.
(C) There is exactly one student in the class for whom the scores of some students are more than 10 marks away.
(D) For each student, the score of exactly one other student is more than 10 marks away.

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The statement has the form $\forall s\ \exists t:\ |score_s - score_t| \leq 10$.
Step 2. Its negation is $\exists s\ \forall t:\ |score_s - score_t| > 10$.
Step 3. This is option (B).

GATE 2026 | Q.6 | 2 marks

Each one of the following clues contains a keyword that is partially filled.
Clue 1: Synonym of recognize (8 letters): _ D _ N T _ F Y
Clue 2: A story long enough to fill a book (5 letters): _ _ _ E L
Clue 3: Two of something (6 letters): _ _ _ P L E
Clue 4: A fraction of something, split equally into two parts (4 letters): _ _ _ F
The first letter of each of the keywords can be rearranged to form a four-letter word. Which one of the options below is a possible choice for the four-letter word?

(A) CHIN
(B) COIN
(C) ITCH
(D) NOSE

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The keywords are IDENTIFY, NOVEL, COUPLE and HALF.
Step 2. Their first letters are I, N, C, H.
Step 3. These can be rearranged to CHIN (COIN needs an O; ITCH needs a T; NOSE needs O and S).

GATE 2026 | Q.7 | 2 marks

Three children P, Q, R and two grown-ups X, Y play a badminton doubles tournament. X and Y are parents to two of the children playing. The child of X is not the same as the child of Y. Exactly one of the children does not have a parent playing in the tournament. The following rules are followed: (i) A parent and his/her child cannot be on the same team. (ii) A match can feature at most one parent and his/her child, that is, a maximum of one parent-child pair can play in a match. The following matches were played: Match 1: P and X vs Q and R; Match 2: P and R vs X and Y; Match 3: R and X vs Q and Y. Which one of the following options is correct?

(A) P does not have any parent playing
(B) Q does not have any parent playing
(C) R does not have any parent playing
(D) X does not have a child playing

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Rule (i): P and X are teammates in Match 1, so P is not X's child; R and X are teammates in Match 3, so R is not X's child. Hence X's child is Q.
Step 2. Q and Y are teammates in Match 3, so Q is not Y's child either, so Y's child is P or R.
Step 3. If R were Y's child, Match 3 would have two parent-child pairs (X–Q and Y–R as opponents): violates rule (ii). So Y's child is P, and R has no parent playing.

GATE 2026 | Q.8 | 2 marks

Let P_k represent the perimeter of a square with sides of length k. The value of the expression $P_1 + P_2 + P_3 + \cdots + P_{10}$ is _____.

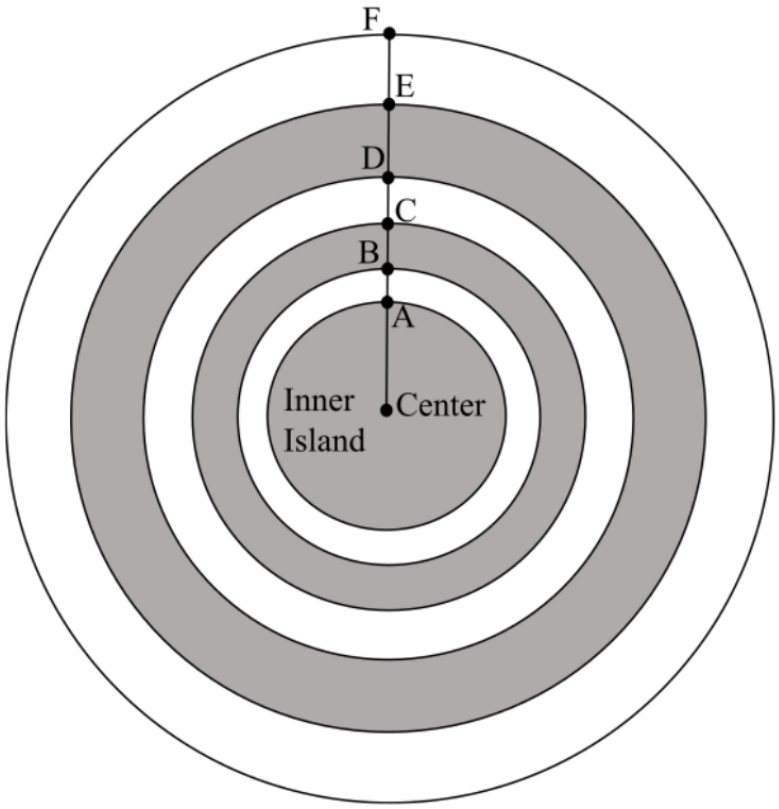
(A) 55
(B) 110
(C) 220
(D) 440

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $P_k = 4k$.
Step 2. $\sum_{k=1}^{10} 4k = 4 \cdot \dfrac{10 \cdot 11}{2} = 220.$

GATE 2026 | Q.9 | 2 marks

The city of Atlantis was crafted by the God of the seas, Poseidon. It was made of alternating concentric circular rings of land (shaded) and water (not shaded) as represented in the figure (not to scale). The radius of Inner Island was 2.5 stades (a unit of length used in ancient Greece). The water surrounding Inner Island was one stade wide (length AB). This was surrounded by two pairs of alternating rings of land and water. The first pair of land and water was two stades wide each (lengths BC and CD), and the outer pair is three stades wide each (lengths DE and EF). The ratio of the surface area of the land to that of the water in the city of Atlantis is _____ (round off to two decimal places).



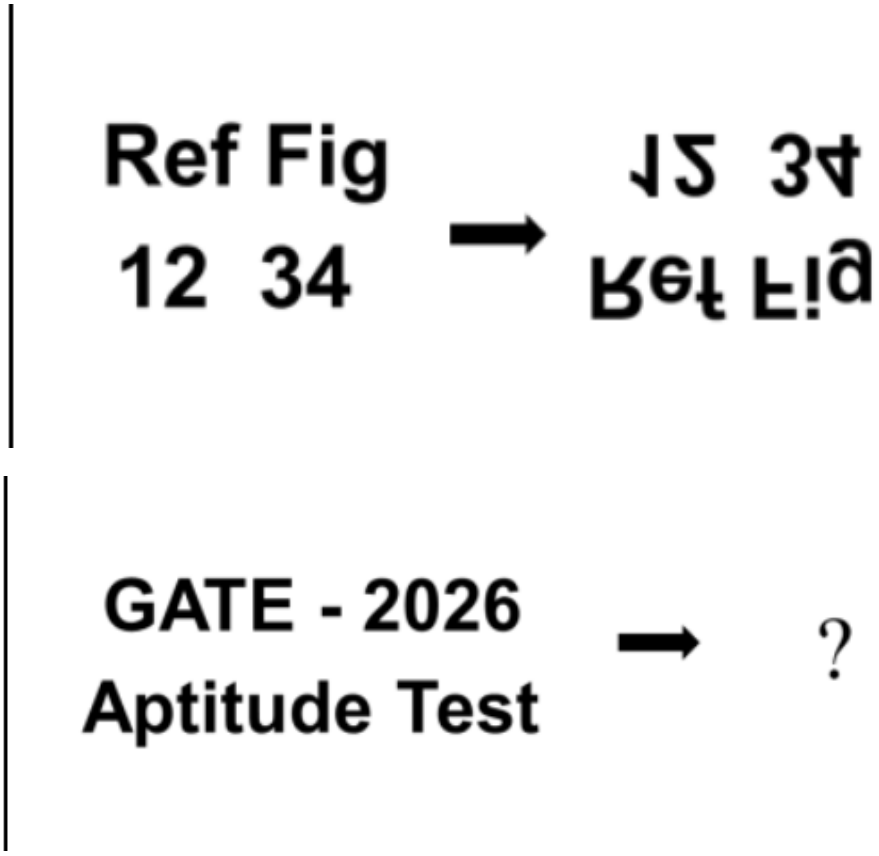
(A) 0.45
(B) 0.60
(C) 0.75
(D) 0.90

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Ring boundaries (radii in stades): island 2.5; water to 3.5; land to 5.5; water to 7.5; land to 10.5; water to 13.5.
Step 2. Area $\propto r^2$. Land: $2.5^2 + (5.5^2 - 3.5^2) + (10.5^2 - 7.5^2) = 6.25 + 18 + 54 = 78.25$.
Step 3. Water: $(3.5^2 - 2.5^2) + (7.5^2 - 5.5^2) + (13.5^2 - 10.5^2) = 6 + 26 + 72 = 104$.
Step 4. Ratio $= \dfrac{78.25}{104} = 0.75$.

GATE 2026 | Q.10 | 2 marks

If [figure] then [figure] (a figure-transformation question; see the figures below). Which one of the options P, Q, R, S is the result?



(A) Test Aptitude
GATE - 2026

(B) Aptitude Test
2026- GATE

(C) Aptitude Test
GATE - 2026

(D) Test Aptitude
2026 - GATE

Answer: (A) [UNVERIFIED: figure hard to read, check against official key]

Solution. Step 1. The transformation shown in the first example is a reflection of the picture about the horizontal axis (vertical flip).
Step 2. Applying the same operation to the second picture gives option P. *(Low confidence.)*

GATE 2026 | Q.11 | 1 mark

Consider the differential equation $\dot{\vec{w}} = A\vec{w}$, with $\vec{w}(t=0) = \begin{bmatrix}1\\1\end{bmatrix}$. If $\vec{w}(t) = e^{t}\vec{u}_x + e^{-2t}\vec{u}_y$ be the solution to the equation where $\vec{u}_x$ and $\vec{u}_y$ are unit vectors along the positive x and y axes respectively, then which of the following options is the correct matrix representing A?

(A) $\begin{bmatrix}1 & 0\\ 0 & -2\end{bmatrix}$

(B) $\begin{bmatrix}-1 & 0\\ 0 & 2\end{bmatrix}$

(C) $\begin{bmatrix}0 & -2\\ 1 & 0\end{bmatrix}$

(D) $\begin{bmatrix}0 & 2\\ -1 & 0\end{bmatrix}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Components: $w_1 = e^{t}$, $w_2 = e^{-2t}$.
Step 2. $\dot{w}_1 = w_1$ and $\dot{w}_2 = -2w_2$.
Step 3. So $A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$.

GATE 2026 | Q.12 | 1 mark

A surface is given by $z^2 = 2x^2 - y^2$ and $\vec{n}$ and $-\vec{n}$ are unit normal vectors to the surface at the point $\vec{P} = \hat{i} + \sqrt{2}\hat{k}$. Which of the following vectors can be $\vec{n}$, where $\hat{i}, \hat{j}$ and $\hat{k}$ are the unit vectors along x, y and z axes, respectively?

(A) $\hat{i} - \sqrt{2}\hat{k}$
(B) $\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}$
(C) $\sqrt{2}\hat{i} - \sqrt{3}\hat{k}$
(D) $\frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Write the surface as $F = 2x^2 - y^2 - z^2 = 0$; the normal is $\nabla F = (4x, -2y, -2z)$.
Step 2. At $(1, 0, \sqrt{2})$: $\nabla F = (4, 0, -2\sqrt{2}) \propto (\sqrt{2}, 0, -1)$.
Step 3. Its unit vector is $\frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}$ (norm $\sqrt{2+1} = \sqrt{3}$): option (D).

GATE 2026 | Q.13 | 1 mark

The Laplace Transform of the signal $x(t) = u(t-2) * (t\,u(t))$ is given by which of the following expressions? ["$*$" represents convolution operator]

(A) $\frac{e^{-2s}}{s^2(s-2)}$
(B) $\frac{e^{-2(s-2)}}{s^3}$
(C) $\frac{s\,e^{-2s}}{(s-2)^2}$
(D) $\frac{e^{-2s}}{s^3}$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Convolution in time is multiplication in the Laplace domain.
Step 2. $\mathcal{L}\{u(t-2)\} = \frac{e^{-2s}}{s}$ and $\mathcal{L}\{t\,u(t)\} = \frac{1}{s^2}$.
Step 3. $X(s) = \frac{e^{-2s}}{s^3}$.

GATE 2026 | Q.14 | 1 mark

Consider carrier transport in a Zener diode in the breakdown region. Which is the dominant transport mechanism for current flow in this case?

(A) Drift
(B) Diffusion
(C) Tunneling
(D) Ballistic transport

Answer: (C) [self-solved, no official key available]

Solution. Step 1. A heavily doped junction has a very thin depletion region, so in reverse breakdown (Zener effect) electrons tunnel directly from the valence band of the p-side into the conduction band of the n-side.

GATE 2026 | Q.15 | 1 mark

Two analog signals $x_1(t)$ and $x_2(t)$ (t in second), are sampled at a rate $F_s = 40$ Hz, where $x_1(t) = \cos(20\pi t)$, $t \geq 0$, and $x_2(t) = \cos(100\pi t)$, $t \geq 0$. The first ten samples (starting from $t = 0$) are considered for the analysis. Which of the following statements is TRUE?

(A) All of the first three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.
(B) All of the last three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.
(C) All of the samples of $x_2(t)$ are greater than the corresponding samples of $x_1(t)$.
(D) All of the fourth to seventh samples of $x_1(t)$ are equal to the corresponding samples of $x_2(t)$.

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Sample at $t = n/40$: $x_1[n] = \cos\left(\frac{\pi n}{2}\right)$ and $x_2[n] = \cos\left(\frac{5\pi n}{2}\right)$.

Step 2. $\frac{5\pi n}{2} = \frac{\pi n}{2} + 2\pi n$, so $x_2[n] = x_1[n]$ for every n (aliasing).

Step 3. Hence the fourth to seventh samples (indeed all samples) are equal: (D).

GATE 2026 | Q.16 | 1 mark

The response of a discrete time system $y[n]$ obeys the following relation: $y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x[n]$. The input to the system is $x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$. Which of the following options is TRUE for $y[n]$?

(A) Stable and causal response
(B) Stable and non-causal response
(C) Unstable and causal response
(D) Unstable and non-causal response

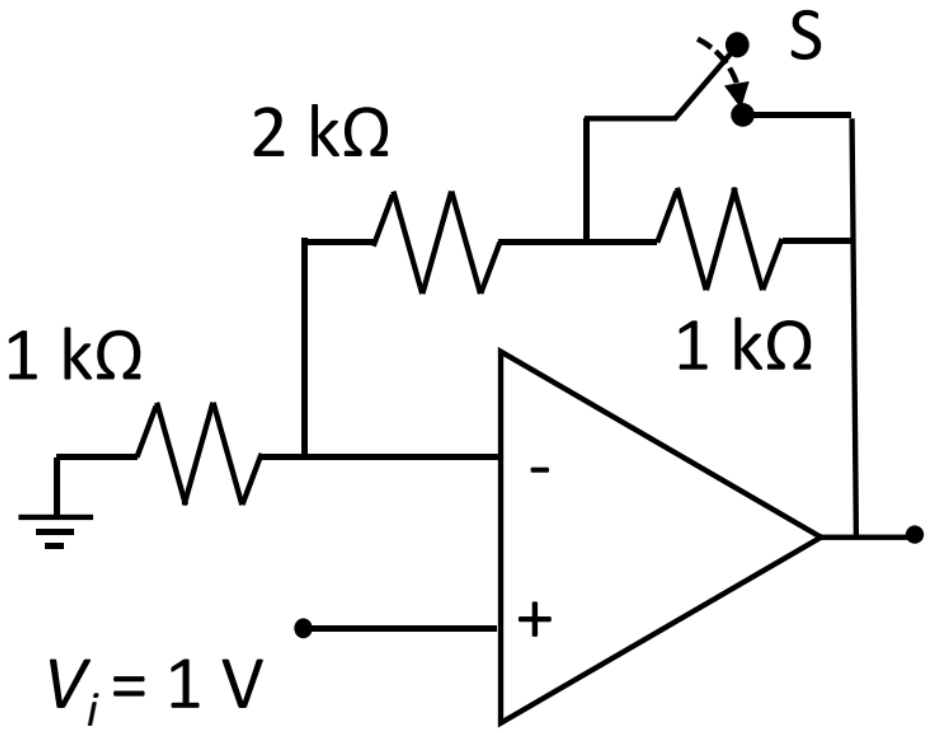
Answer: (A) [self-solved, no official key available]

Solution. Step 1. Characteristic equation: $z^2 - \frac{5}{6}z + \frac{1}{6} = 0 \Rightarrow z = \frac{1}{2}, \frac{1}{3}$.

Step 2. Both poles lie inside the unit circle and the recursion is causal, so the response is stable and causal.

GATE 2026 | Q.17 | 1 mark

The ideal OP-AMP circuit shown in the Figure produces output voltage $V_o = x$ when the Switch, S, is open. Which of the options represents the output voltage when S is closed?



(A) x
(B) $\frac{2}{3}x$
(C) $\frac{3}{4}x$
(D) $\frac{1}{2}x$

Answer: (C) [self-solved, no official key available]

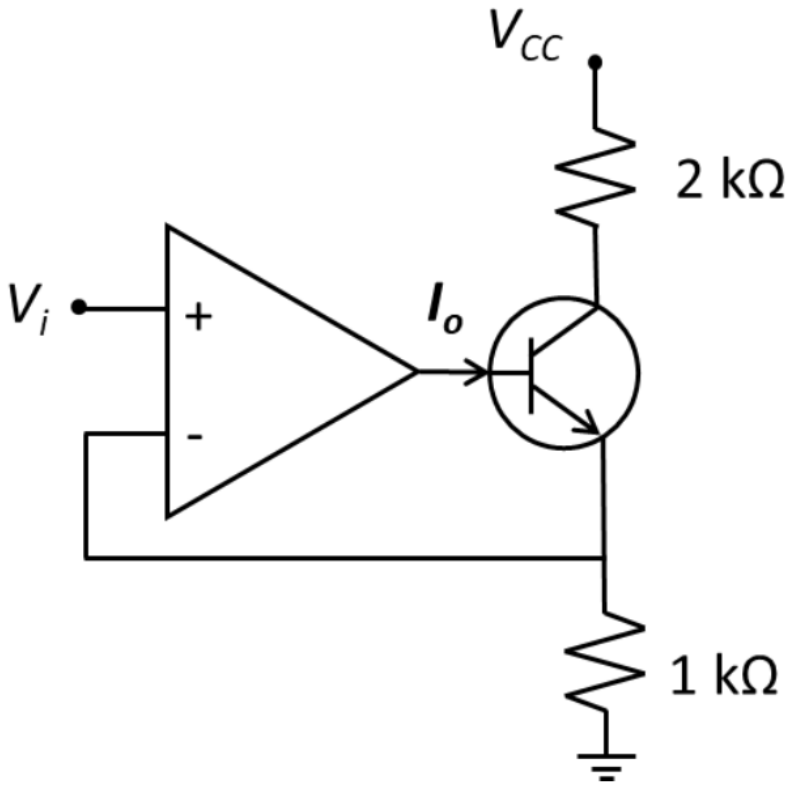
Solution. Step 1. The circuit is a non-inverting amplifier: $V_o = \left(1 + \frac{R_f}{1\text{k}}\right) V_i$.

Step 2. Switch open: $R_f = 2\text{k} + 1\text{k} = 3$ kΩ, $x = (1+3)(1) = 4$ V.

Step 3. Switch closed (shorts the 1 kΩ in the feedback path): $R_f = 2$ kΩ, $V_o = 3\text{ V} = \frac{3}{4}x$.

GATE 2026 | Q.18 | 1 mark

Consider the circuit shown in the Figure with $V_i = 3$ V and $V_{CC} = 12$ V. Assume $V_{BE} = 0.7$ V and $\beta_{dc} = 99$ for the BJT. Which of the following options is the correct value of the current I_o?



(A) 60 μA
(B) 6 μA
(C) 3 μA
(D) 30 μA

Answer: (D) [self-solved, no official key available]

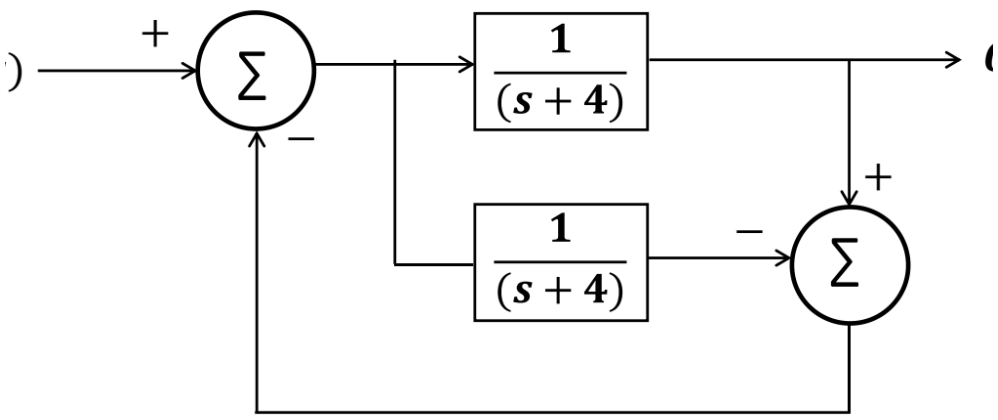
Solution. Step 1. The op-amp forces the emitter (feedback) node to equal V_i: $V_E = 3$ V.

Step 2. $I_E = \dfrac{3\text{ V}}{1\text{ k}\Omega} = 3$ mA.

Step 3. The op-amp supplies the base current: $I_o = I_B = \dfrac{I_E}{\beta + 1} = \dfrac{3\text{ mA}}{100} = 30\ \mu$A.

GATE 2026 | Q.19 | 1 mark

A control system is shown in the Figure. Which option represents the correct transfer function of the system?



(A) $\dfrac{1}{(s+4)^2}$

(B) $\dfrac{1}{s+4}$

(C) $\dfrac{2}{s+4}$

(D) $\dfrac{1}{s^2+8s+17}$

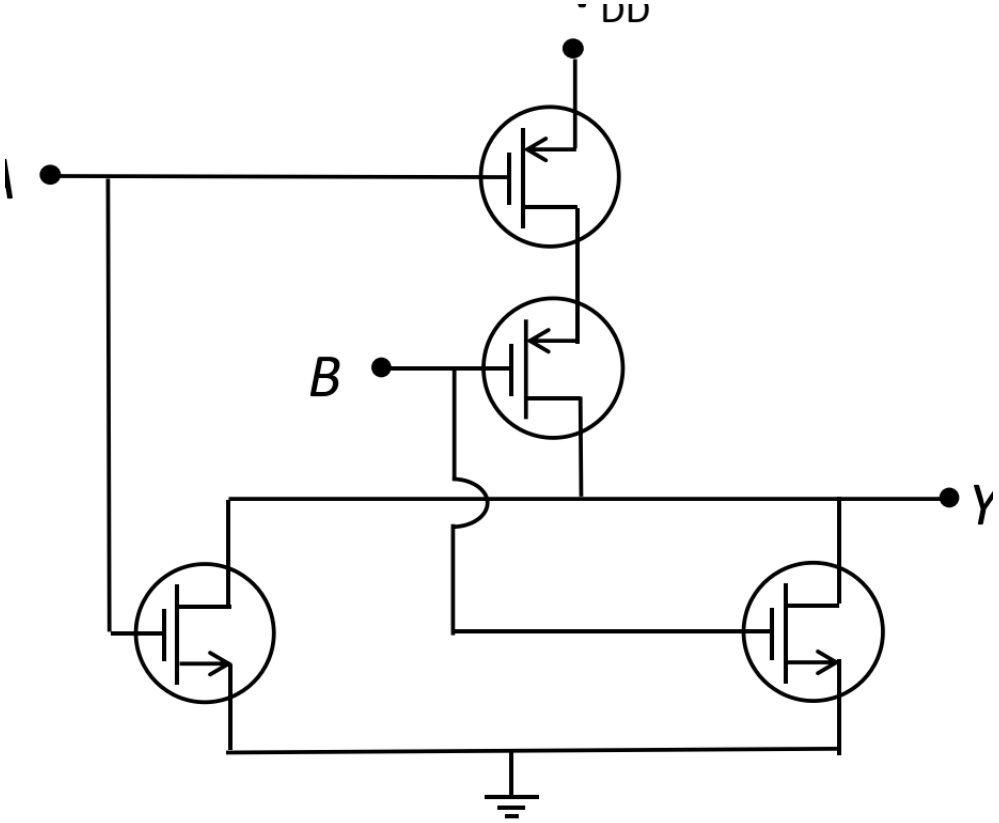
Answer: (D) [self-solved, no official key available]

Solution. Step 1. The two blocks $\dfrac{1}{s+4}$ in cascade with unity negative feedback give $G(s) = \dfrac{1}{(s+4)^2}$ in the loop.

Step 2. $\dfrac{C}{R} = \dfrac{G}{1+G} = \dfrac{1}{(s+4)^2+1} = \dfrac{1}{s^2+8s+17}$.

GATE 2026 | Q.20 | 1 mark

In the circuit shown in the Figure, A and B are logic inputs and Y is the logic output. Which of the following logic operations is realized by the circuit?



(A) NOR
(B) OR
(C) XOR
(D) AND

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Two pMOS transistors in series connect V_{DD} to Y; two nMOS transistors in parallel connect Y to ground.
Step 2. $Y = 1$ only when both inputs are 0 (both pMOS on, both nMOS off). That is the NOR function.

GATE 2026 | Q.21 | 1 mark

Consider the Friis' transmission equation $P_R = \dfrac{P_T G_T G_R \lambda^2}{(4\pi D)^2}$, where P_R and P_T are the received and the transmitted powers, respectively. G_T and G_R are the gain of transmitting and receiving antennas, respectively, D is the distance between the transmitting and receiving antennas, and λ is the wavelength in free space. Given: $G_T = G_R = 1.0$, $\lambda = 0.30$ m and $P_T = +10$ dBm. Choose the distance (D), in km, from the following options at which the received power, $P_R = -90$ dBm?

(A) $\dfrac{15}{4\pi}$
(B) $\dfrac{15}{2\pi}$
(C) $\dfrac{75}{2\pi}$
(D) $\dfrac{3}{4\pi}$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $\dfrac{P_R}{P_T} = -100$ dB $= 10^{-10}$.

Step 2. $\dfrac{\lambda^2}{(4\pi D)^2} = 10^{-10} \Rightarrow 4\pi D = \lambda \times 10^5 = 3 \times 10^4$ m.

Step 3. $D = \dfrac{3 \times 10^4}{4\pi}$ m $= \dfrac{30}{4\pi}$ km $= \dfrac{15}{2\pi}$ km.

GATE 2026 | Q.22 | 1 mark

Consider a discrete memoryless source with an alphabet of four source symbols. $s(t)$ is a multi-level $(-1, 0, +1, +2)$ signal representing a long sequence of random symbols from the above source which is generating 10^4 symbols per second. Which of the following options is the correct value of equivalent Nyquist bandwidth of $s(t)$?

(A) 10 kHz
(B) 64 kHz
(C) 5 kHz
(D) 20 kHz

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The signalling rate is $R_s = 10^4$ symbols/s.

Step 2. The minimum (Nyquist) bandwidth for a multilevel pulse train is $B = \dfrac{R_s}{2} = 5$ kHz (the number of levels does not change it).

GATE 2026 | Q.23 | 1 mark

The relation between the input current (I) and the output voltage (V) of a circuit is governed by the equation: $C\dfrac{dV}{dt} = I(t) - m(t)$. The circuit is excited by $I(t) = q\,\delta(t)$, where q is a real valued constant. V at $t = 0^-$ is V_0. Which of the following is an equivalent representation of the above case?

(A) $C\dfrac{dV}{dt} = -m(t)$, with $V(t = 0^-) = V_0 + q/C$
(B) $C\dfrac{dV}{dt} = -m(t)$, with $V(t = 0^-) = V_0 + q/C + m(t = 0^-)$
(C) $C\dfrac{dV}{dt} = -m(t)$, with $V(t = 0^-) = V_0 - q/C + m(t = 0^-)$
(D) $C\dfrac{dV}{dt} = -m(t)$, with $V(t = 0^-) = q/C$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. An impulse of charge q applied to a capacitor makes the voltage jump instantly by q/C.
Step 2. The impulse can therefore be replaced by an equivalent initial condition $V_0 + q/C$ with the homogeneous equation $C\dot{V} = -m(t)$.

GATE 2026 | Q.24 | 1 mark

The electric field of a monochromatic plane wave travelling in a lossless isotropic and homogeneous medium is given by $\vec{E}(z,t) = E_0\,[\hat{x}\cos(\omega t - kz) + \hat{y}\sin(\omega t - kz)]$ in a right-handed orthogonal co-ordinate system. Which of the following is the correct polarization of the electromagnetic wave?

(A) Right-handed circularly polarized
(B) Left-handed circularly polarized
(C) Linearly polarized
(D) Linearly polarized with -45° angle to $\hat{x}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The two components have equal amplitudes and a 90° phase difference: circular polarization.
Step 2. At $z = 0$: $\vec{E} = E_0(\hat{x}\cos\omega t + \hat{y}\sin\omega t)$ rotates from $\hat{x}$ towards $\hat{y}$ (counter-clockwise when viewed against the direction of propagation $+z$), i.e. right-hand rotation about the propagation direction: right-handed circular

polarization.

GATE 2026 | Q.25 | 1 mark

Consider a p-n junction diode when it is forward biased with 2 V. Which of the following is/are the correct magnitude(s) of the energy difference between quasi Fermi-levels, E_{fn} in the n-side and E_{fp} in the p-side?

(A) 2 eV
(B) 1 eV
(C) 2 V
(D) 1 V

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The separation of the quasi-Fermi levels equals qV_a.
Step 2. For $V_a = 2$ V this is 2 eV.

GATE 2026 | Q.26 | 1 mark

Consider the matrix $M = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 0 \\ -1 & a & b \end{bmatrix}$. Which of the following options is/are TRUE if $\det(M) \neq 0$?

(A) $a = -\frac{1}{2}$ and $b = -\frac{1}{2}$
(B) $a = \frac{1}{2}$ and $b = \frac{1}{2}$
(C) $a = -3$ and $b = 0$
(D) $a = \frac{1}{2}$ and $b = -3$

Answer: (B), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. Expand along the first row: $\det M = 2(3b - 0) - 1(b - 0) + 1(a + 3) = 5b + a + 3$.
Step 2. (A): $-2.5 - 0.5 + 3 = 0$; (C): $0 - 3 + 3 = 0$ (singular).
Step 3. (B): $2.5 + 0.5 + 3 = 6 \neq 0$; (D): $-15 + 0.5 + 3 = -11.5 \neq 0$.

GATE 2026 | Q.27 | 1 mark

A binary ripple counter is designed to count $(0)_{10}$ to $(64)_{10}$. Which of the following is/are the number of flip-flops required to design the counter?

(A) 6
(B) 7
(C) 4
(D) 5

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Counting from 0 to 64 requires 65 distinct states.
Step 2. $2^6 = 64 < 65 \leq 2^7$, so 7 flip-flops are needed.

GATE 2026 | Q.28 | 1 mark

Which option(s) represents/represent the dielectric loss tangent of a substrate?

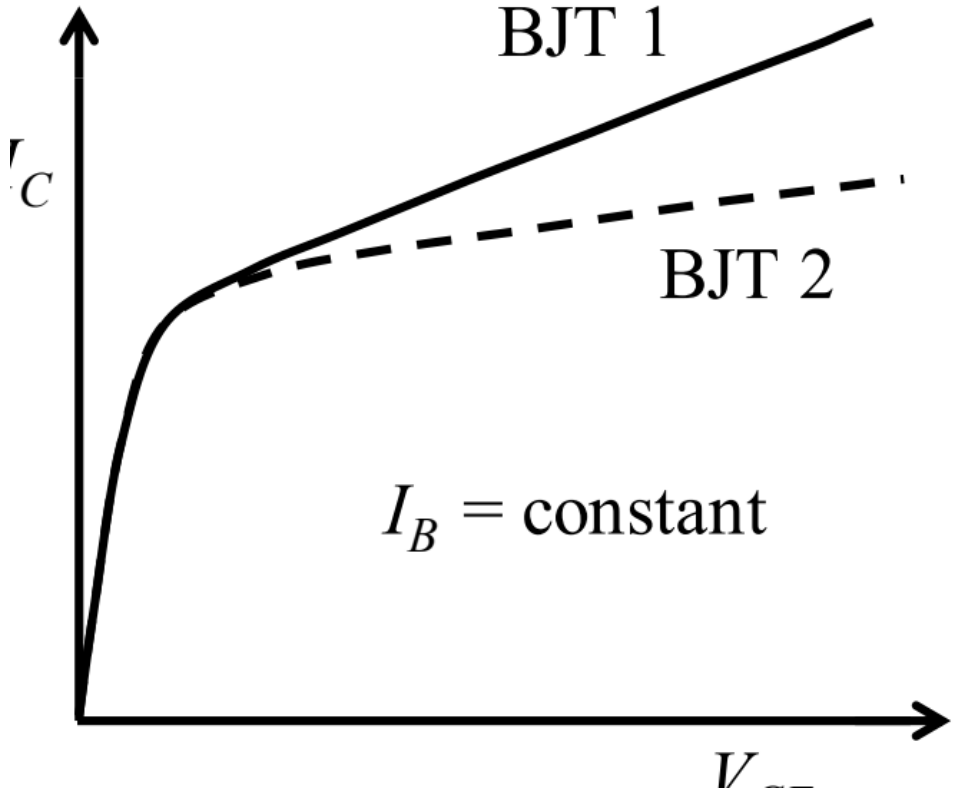
(A) Ratio of the real to imaginary parts of the total displacement current
(B) $(\omega\epsilon'' + \sigma)/(\omega\epsilon')$
(C) Ratio of the electric susceptibility to permittivity
(D) Ratio of the polarization vector $\vec{P}$ to the displacement vector $\vec{D}$

Answer: (A), (B) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. The loss tangent is $\tan\delta = \dfrac{\omega\epsilon'' + \sigma}{\omega\epsilon'}$ (B).
Step 2. This equals the ratio of the in-phase (loss) to quadrature (displacement) parts of the total current (A).
Step 3. (C) and (D) are not loss ratios.

GATE 2026 | Q.29 | 1 mark

Figure shows the output characteristics of two different Bipolar Junction Transistors (BJT), BJT 1 with magnitude of Early voltage $|V_{A1}|$, and BJT 2 with magnitude of Early voltage $|V_{A2}|$. Which of the following options is/are correct regarding the Early voltages?



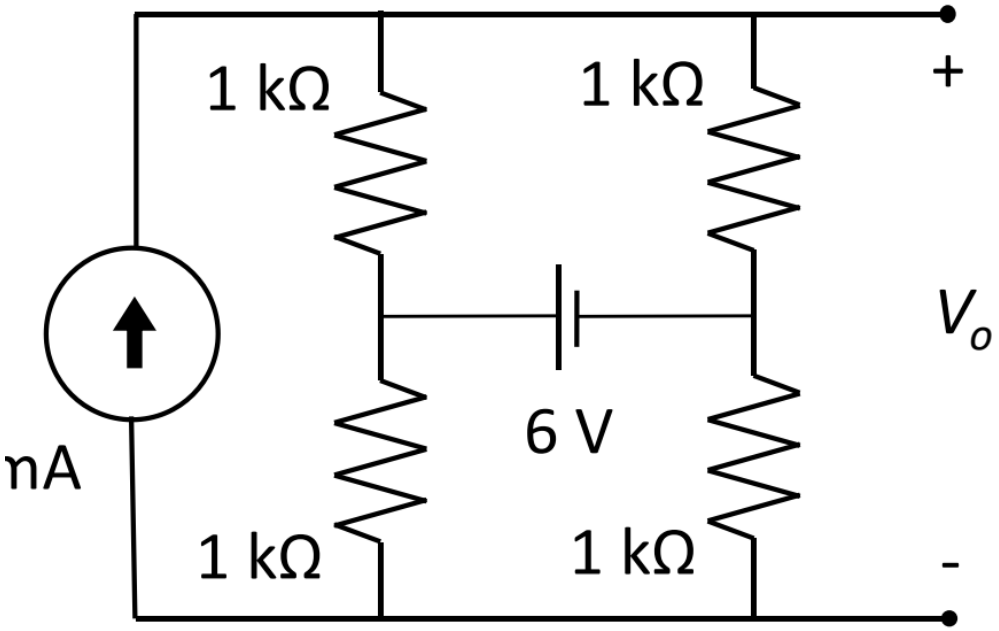
(A) $|V_{A1}| > |V_{A2}|$
(B) $|V_{A1}|$ is infinitely large
(C) $|V_{A1}| < |V_{A2}|$
(D) $|V_{A2}|$ is finite

Answer: (A), (D) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. A flatter I_C–V_{CE} curve corresponds to a larger Early voltage (I_C slope $= I_C/V_A$).
Step 2. BJT 1 is flatter than BJT 2, so $|V_{A1}| > |V_{A2}|$ (A); BJT 2 has a visible slope, so V_{A2} is finite (D). BJT 1 still shows a slight slope, so it is not infinite.

GATE 2026 | Q.30 | 1 mark

The output voltage V_o (in Volt) for the network given in the Figure is _____ (rounded off to two decimal places).



Answer: 2.33 [UNVERIFIED: figure hard to read, check against official key]

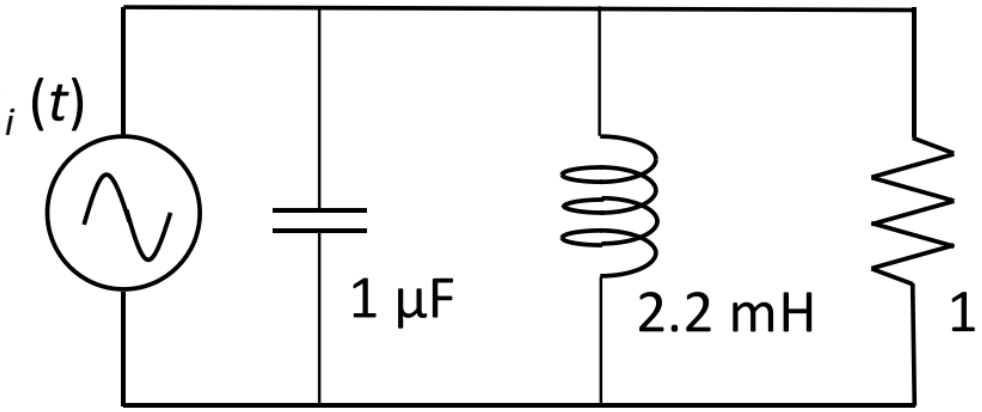
Solution. Step 1. Work in kΩ, mA, V. Let V_2 be the voltage at the junction fed by the 1 mA source through the first 1 kΩ.

Step 2. KCL: $1 = (V_2 - 6) + \dfrac{V_2}{2} \Rightarrow 1.5V_2 = 7 \Rightarrow V_2 = 4.67$ V.

Step 3. V_o is the drop across one of the two series 1 kΩ resistors: $V_o = \dfrac{V_2}{2} = 2.33$ V. *(Low confidence: circuit read from a scan.)*

GATE 2026 | Q.31 | 1 mark

Consider the circuit shown in the Figure, where the input $v_i(t)$ is in Volt. The average power (in mW) dissipated in the load resistance of 1 kΩ at the resonant frequency is _____ (rounded off to two decimal places).



Answer: 72.00 [self-solved, no official key available]

Solution. Step 1. At resonance the series L–C combination has zero net reactance, so the full source voltage appears across the load: $V_m = 12$ V.

Step 2. $P = \dfrac{V_m^2}{2R} = \dfrac{144}{2 \times 1000} = 72$ mW.

GATE 2026 | Q.32 | 1 mark

A wireless digital transmission scheme is using 16-QAM over an additive white Gaussian noise channel and a maximum-likelihood receiver. Consider the information bit rate from source to be 4×10^6 bits per second. The minimum transmission bandwidth (in MHz) of the modulated signal necessary for optimum recovery of information at the receiver is _____ (rounded off to two decimal places).

Answer: 1.00 [self-solved, no official key available]

Solution. Step 1. 16-QAM carries $\log_2 16 = 4$ bits per symbol, so the symbol rate is $\dfrac{4 \times 10^6}{4} = 10^6$ symbols/s.
Step 2. The minimum bandwidth of the modulated (pass-band) signal equals the symbol rate: 1.00 MHz.

GATE 2026 | Q.33 | 1 mark

The cutoff frequency (in GHz) for the dominant TE_{10} mode of an air-filled rectangular waveguide of inner dimension 0.28 inch $\times$ 0.14 inch is _____ (rounded off to two decimal places).

Answer: 21.09 [self-solved, no official key available]

Solution. Step 1. $a = 0.28$ in $= 7.112$ mm.
Step 2. $f_c = \dfrac{c}{2a} = \dfrac{3 \times 10^8}{2 \times 7.112 \times 10^{-3}} = 21.09$ GHz.

GATE 2026 | Q.34 | 1 mark

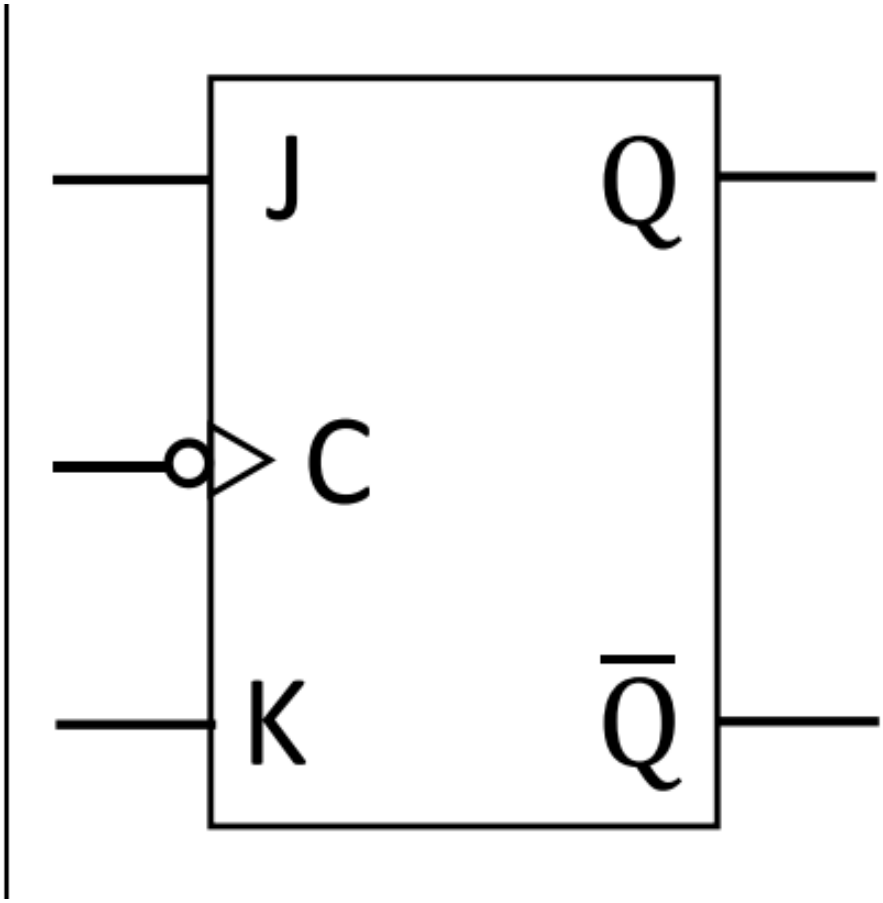
For a lossless passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at -3 dB. For a lossy passive two-port network, $|S_{11}|$ and $|S_{21}|$ intersect at -4 dB. The percentage of power dissipated in the lossy network at the intersection frequency is _____ (rounded off to two decimal places).

Answer: 20.38 [self-solved, no official key available]

Solution. Step 1. At the crossing $|S_{11}|^2 = |S_{21}|^2 = 10^{-0.4} = 0.398$.
Step 2. Reflected plus transmitted power $= 0.796$.
Step 3. Dissipated power fraction $= 1 - 0.796 = 0.204 = 20.38\%$.

GATE 2026 | Q.35 | 1 mark

The negative edge triggered JK flip-flop in the Figure has J and K inputs tied to Logic High and a square wave of 10 cycles/second is applied to its clock (C) input. The frequency of the output Q (in cycles/second) is _____ (rounded off to two decimal places).



Answer: 5.00 [self-solved, no official key available]

Solution. Step 1. With $J = K = 1$ the flip-flop toggles on every falling clock edge.
Step 2. The output frequency is half the clock: 5.00 Hz.

GATE 2026 | Q.36 | 2 marks

Consider the two series, S_A and S_B, where $S_A = \sum_{n=1}^{\infty} \frac{n^2}{2^n}$, $S_B = 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} + \frac{1}{128} + \frac{1}{512} + \cdots$. Which of the following statements is correct for the two given series?

(A) Both S_A and S_B converge.
(B) Neither S_A nor S_B converges.
(C) S_A converges but S_B does not converge.
(D) S_B converges but S_A does not converge.

Answer: (A) [self-solved, no official key available]

Solution. Step 1. S_A: ratio test $\frac{(n+1)^2/2^{n+1}}{n^2/2^n} \to \frac{1}{2} < 1$: converges (to 6).

Step 2. S_B consists of terms 2^{-k} with $k = 0, 1, 3, 4, 6, 7, 9, \ldots$, a subset of the terms of $\sum_{k \geq 0} 2^{-k} = 2$; being a series of positive terms bounded above by 2, it converges.

GATE 2026 | Q.37 | 2 marks

The continuous time signal $x(t)$ is real, periodic with period T and satisfies the Dirichlet conditions. The Fourier series representation is $x(t) = \sum_{n=-\infty}^{\infty} a_n e^{j\frac{2\pi nt}{T}}$ and $x(t)$ satisfies the following: $x\left(t - \frac{T}{2}\right) = -x(t)$. For any integer m, which of the following options is correct?

(A) $a_{2m} = 0$
(B) $a_{2m} = 1$
(C) $a_{2m} = a_{2m+1}$
(D) $a_{2m} = -1$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. A delay of $T/2$ multiplies the n-th coefficient by $e^{-j\pi n} = (-1)^n$.
Step 2. $x(t - T/2) = -x(t)$ requires $(-1)^n a_n = -a_n$, i.e. $a_n = 0$ for even n.

GATE 2026 | Q.38 | 2 marks

Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows Normal distribution with zero mean and unity variance. Y and Z are defined as $Y = X + N$ and $Z = X^2 + N$. Which of the following pairs represents the values of correlation between X and Y and that between X and Z?

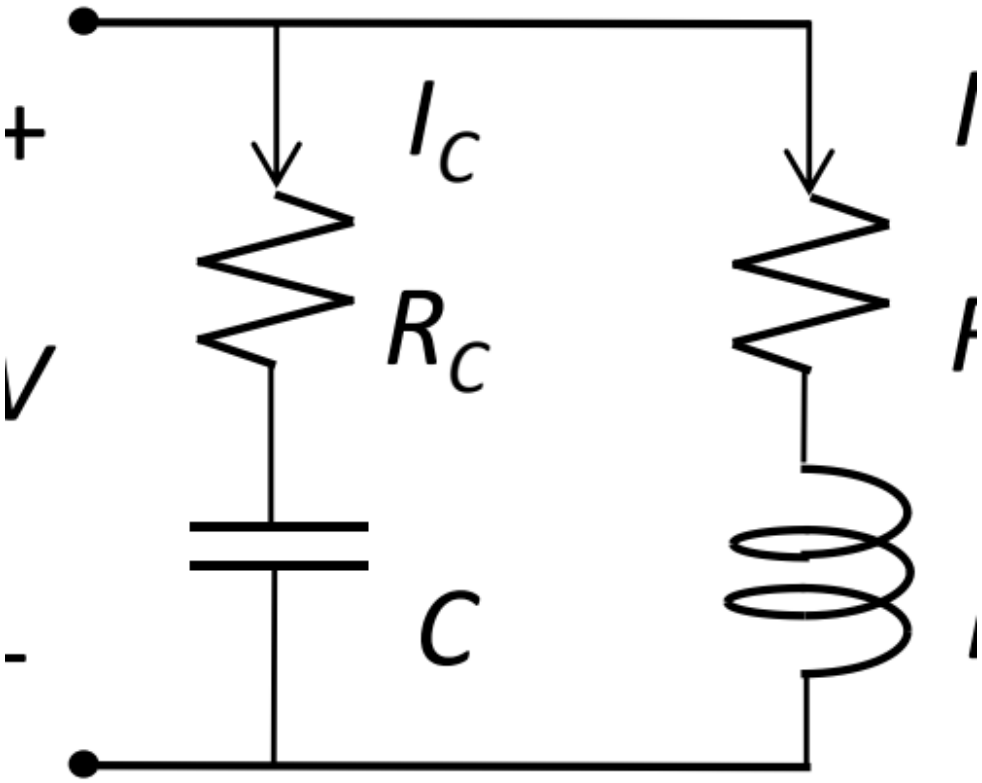
(A) 1/3 and 0
(B) 1/3 and 1/9
(C) 1/3 and 1/3
(D) 1 and 0

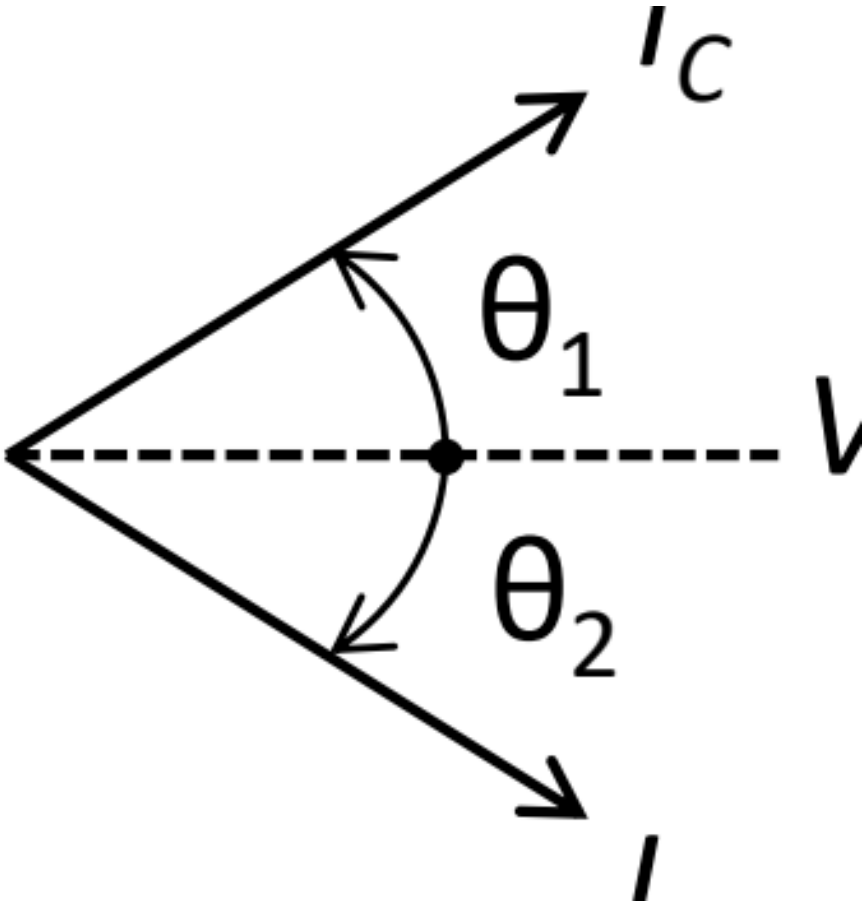
Answer: (A) [self-solved, no official key available]

Solution. Step 1. $E[XY] = E[X^2] + E[X]E[N] = \frac{1}{3} + 0 = \frac{1}{3}$.
Step 2. $E[XZ] = E[X^3] + E[X]E[N] = 0$ because X is symmetric.

GATE 2026 | Q.39 | 2 marks

In the given circuit, $L = 1$ μH and $C = 1$ μF. The phasor diagram for I_C and I_L is also shown. Assume that the phase $(\theta_1 + \theta_2)$ is 90° at a frequency of 159.15 kHz. Among the following options, what is the nearest integer value of $R_C \times R_L$?





(A) 0

(B) 1
(C) 2
(D) 10

Answer: (B) [self-solved, no official key available]

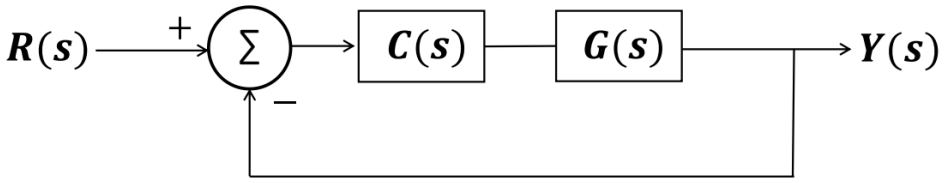
Solution. Step 1. $\omega = 2\pi(159.15\text{ kHz}) = 10^6$ rad/s $= \dfrac{1}{\sqrt{LC}}$.

Step 2. $\tan\theta_1 = \dfrac{1}{\omega C R_C}$ and $\tan\theta_2 = \dfrac{\omega L}{R_L}$. $\theta_1 + \theta_2 = 90^\circ \Rightarrow \tan\theta_1 \tan\theta_2 = 1$.

Step 3. $\dfrac{L}{CR_CR_L} = 1 \Rightarrow R_C R_L = \dfrac{L}{C} = 1\ \Omega^2$.

GATE 2026 | Q.40 | 2 marks

For the control system shown in the Figure, the transfer function of a plant, $G(s) = \dfrac{1}{(s+1)(s+2)}$ is connected in cascade with a compensator $C(s) = K(s+\alpha)$, where K and α are positive real valued constants. Which of the following pairs (K, α) represent the correct values for the closed loop system to have poles at $(-3 \pm j\sqrt{5})$?



(A) 2, 3
(B) 3, 4
(C) 2, 4
(D) 3, 3

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Characteristic equation: $1+K(s+\alpha)G(s) = 0 \Rightarrow s^2+3s+2+K(s+\alpha) = s^2+(3+K)s+(2+K\alpha) = 0$.
Step 2. Poles $-3 \pm j\sqrt{5}$: $s^2 + 6s + 14$. Matching: $3 + K = 6 \Rightarrow K = 3$; $2 + K\alpha = 14 \Rightarrow \alpha = 4$.

GATE 2026 | Q.41 | 2 marks

The state and output equations for a control system are: $\dot{\mathbf{x}} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} \mathbf{x}$.

Which of the following expressions correctly represents the transfer function $\dfrac{Y(s)}{U(s)}$ of the system with zero initial conditions?

(A) $\dfrac{3s}{s^2 + 4s - 6}$
(B) $\dfrac{3s + 5}{s^2 + 4s - 6}$
(C) $\dfrac{3s + 5}{s^2 + 4s + 6}$
(D) $\dfrac{3s}{s^2 + 4s + 6}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $(sI - A) = \begin{bmatrix} s+4 & 1.5 \\ -4 & s \end{bmatrix}$, $\det = s^2 + 4s + 6$.

Step 2. $(sI-A)^{-1}B = \dfrac{1}{s^2+4s+6}\begin{bmatrix} s & -1.5 \\ 4 & s+4 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \dfrac{1}{s^2+4s+6}\begin{bmatrix} 2s \\ 8 \end{bmatrix}$.

Step 3. $H(s) = \dfrac{1.5(2s)+0.625(8)}{s^2+4s+6} = \dfrac{3s+5}{s^2+4s+6}$.

GATE 2026 | Q.42 | 2 marks

The address of the first location of a 256 kilo byte (KB) memory is $(2500)_H$. Choose the correct address of the last location of the memory.

(A) $(2FFF)_H$
(B) $(124FF)_H$
(C) $(424FF)_H$
(D) $(324FF)_H$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. 256 KB $= 2^{18}$ locations $= 40000_H$.
Step 2. Last address $= 2500_H + 40000_H - 1 = 424FF_H$.

GATE 2026 | Q.43 | 2 marks

Consider a real signal $x(t)$, $-\infty < t < \infty$, such that $x(t) = 0$ for $t < 0$, $x(t) = 2$ for $0 \le t < 1$ and $x(t) = 0$ for $t \ge 1$. Let $E[x(t)] = \int_{-\infty}^{\infty}[x(t)]^2 dt$. Which of the following options correctly represents the ratio $\dfrac{E[x(t)]}{E[3x(-3t+5)]}$?

(A) 3
(B) 1
(C) 1/3
(D) 1/9

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $E[x(t)] = 2^2 \times 1 = 4$.
Step 2. $3x(-3t+5)$ is non-zero when $0 \le -3t+5 < 1$, i.e. over a width of 1/3, with amplitude 6: energy $= 36 \times \dfrac{1}{3} = 12$.
Step 3. Ratio $= \dfrac{4}{12} = \dfrac{1}{3}$.

GATE 2026 | Q.44 | 2 marks

A QPSK modulated signal from an additive white Gaussian noise (AWGN) channel is received with an $E_b/N_0 = 8.4$ dB at the input of a coherent QPSK demodulator. A maximum-likelihood reception method is used in the demodulator. Assume the complementary error function $\text{erfc}(u) \simeq \dfrac{1}{u\sqrt{\pi}}\exp(-u^2)$. Which is the nearest bit error rate (BER) at the output of the demodulator?

(A) 10^{-3}
(B) 10^{-4}
(C) 10^{-5}
(D) 10^{-6}

Answer: (B) [self-solved, no official key available]

Solution. Step 1. QPSK: BER $= \dfrac{1}{2}\,\text{erfc}\left(\sqrt{E_b/N_0}\right)$.
Step 2. $E_b/N_0 = 10^{0.84} = 6.92$, $u = \sqrt{6.92} = 2.63$.
Step 3. BER $\approx \dfrac{1}{2}\cdot\dfrac{e^{-6.92}}{2.63\sqrt{\pi}} = \dfrac{1}{2}\cdot\dfrac{9.9\times10^{-4}}{4.66} = 1.1\times10^{-4} \approx 10^{-4}$.

GATE 2026 | Q.45 | 2 marks

What is the 10's complement of $(47)_{10}$?

(A) 52
(B) 53
(C) 54
(D) 55

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The 10's complement of an n-digit number N is $10^n - N$.
Step 2. $100 - 47 = 53$.

GATE 2026 | Q.46 | 2 marks

Consider a real baseband signal $x(t) = e^{-2t}$, for t (in seconds) ≥ 0. If 99% of energy of $x(t)$ lies within B Hz, then which of the following options is TRUE for the value of B?

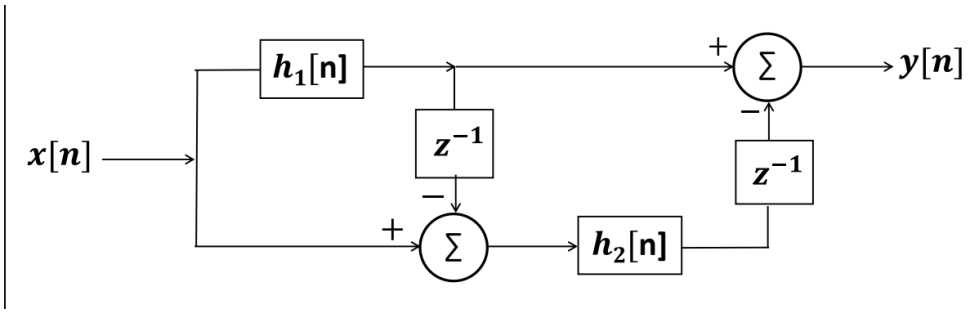
(A) $B > 1$ kHz
(B) $63/\pi$ Hz $< B < 64/\pi$ Hz
(C) $126/\pi$ Hz $< B < 128/\pi$ Hz
(D) $B < 1$ Hz

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Total energy $E = \int_0^\infty e^{-4t} dt = \frac{1}{4}$.
Step 2. $|X(f)|^2 = \frac{1}{4 + 4\pi^2 f^2}$; the energy within $|f| < B$ is $\frac{1}{2\pi} \arctan(\pi B)$.
Step 3. Set $\frac{1}{2\pi} \arctan(\pi B) = 0.99 \cdot \frac{1}{4}\cdot$ (normalised) $\Rightarrow \frac{2}{\pi} \arctan(\pi B) = 0.99$, so $\pi B = \tan(0.99\pi/2) = 63.66$.
Step 4. $B = \frac{63.66}{\pi}$ Hz: $\frac{63}{\pi} < B < \frac{64}{\pi}$.

GATE 2026 | Q.47 | 2 marks

Consider the discrete time system (S) with input $x[n]$ and output $y[n]$ as shown in the Figure. The two sub-systems represented by their impulse responses $h_1[n]$ and $h_2[n]$ are linear and time invariant. Which of the following statements is necessarily TRUE?



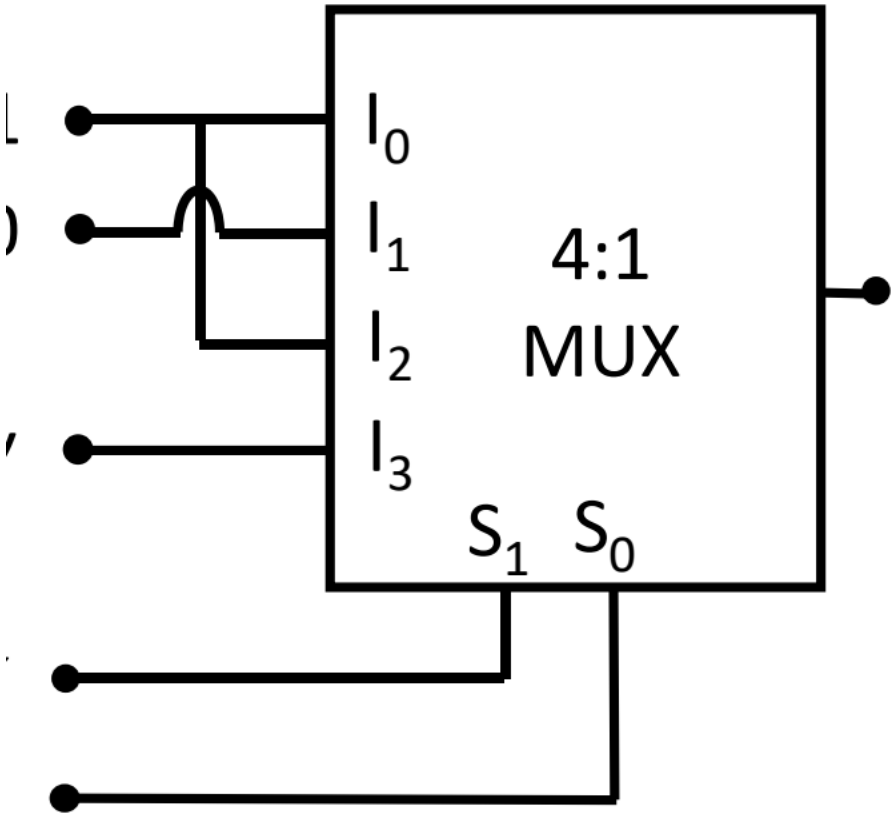
(A) S is causal.
(B) S is linear and time invariant.
(C) S is linear and time varying.
(D) S is non-linear.

Answer: (B) [self-solved, no official key available]

Solution. Step 1. S is built only from LTI blocks, delays and adders (summers).
Step 2. Such interconnections are linear and time invariant. Causality depends on whether h_1, h_2 are causal, so it is not guaranteed.

GATE 2026 | Q.48 | 2 marks

A Boolean function, $f(x, y, z)$ with x as MSB and z as LSB is realized by 4:1 multiplexer (MUX) with select lines, S_1 and S_0 (S_1 is MSB, S_0 is LSB) and inputs, I_0, I_1, I_2, I_3 as shown in the Figure. Which of the following options is the correct expression of $f(x, y, z)$?



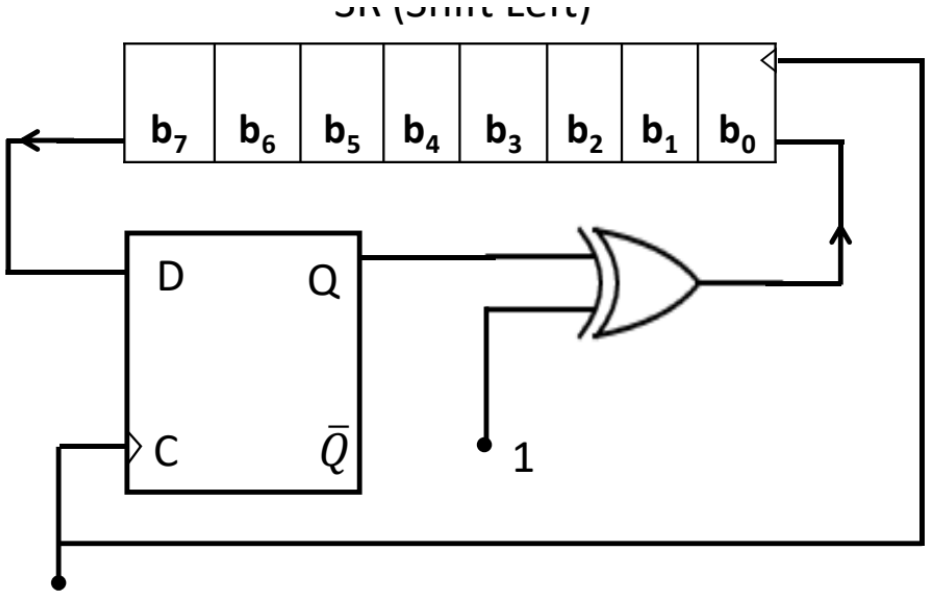
(A) $xz + y$
(B) $x\bar{y} + z$
(C) $xy + \bar{z}$
(D) $\bar{x}y + \bar{z}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Select lines: $S_1 = x$, $S_0 = z$. Inputs: $I_0 = 1$, $I_1 = 0$, $I_2 = 1$, $I_3 = y$.
Step 2. $f = \bar{x}\bar{z} \cdot 1 + \bar{x}z \cdot 0 + x\bar{z} \cdot 1 + xz \cdot y = \bar{z} + xyz \cdot \ldots = \bar{z} + xy$.

GATE 2026 | Q.49 | 2 marks

A shift-left Shift Register (SR) and a D flip-flop are connected to a synchronized clock as shown in the Figure. Assume that the SR and D flip-flops are initially cleared and the XOR gate has no propagation delay. Which of the following options gives the correct binary representation ($b_7b_6b_5b_4b_3b_2b_1b_0$) of the content of the shift register immediately after the 5th clock transition (positive edge)?



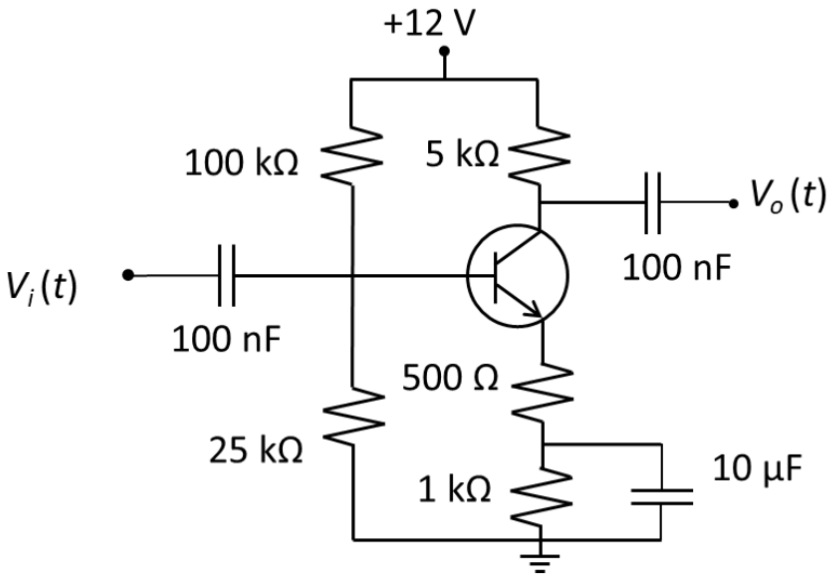
(A) 00011111
(B) 10111111
(C) 00111111
(D) 11000011

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The serial input to the register is the XOR of the register output b_7 with the flip-flop output; both are 0 initially, and the flip-flop supplies a 1 each clock.
Step 2. A logic 1 is shifted in at each clock; after 5 clocks the register holds 00011111.

GATE 2026 | Q.50 | 2 marks

A small signal source, $V_i(t) = A\cos(10^5 t) + B\sin(10^7 t)$ is applied to a BJT circuit as shown in the Figure. Assume zero source resistance, $V_{BE} = 0.7$ V, $\beta_{dc} = 99$, Early voltage = 100 V and Thermal voltage = 25 mV. Effect of internal parasitic capacitances of the BJT may be neglected. Which expression is the best approximation of the output voltage $V_o(t)$?



(A) $-9.1\left[A\cos(10^5t)+B\sin(10^7t)\right]$
(B) $9.1\left[A\cos(10^5t)-B\sin(10^7t)\right]$
(C) $-190.4\left[A\cos(10^5t)+B\sin(10^7t)\right]$
(D) $190.4\left[A\cos(10^5t)-B\sin(10^7t)\right]$

Answer: (A) [self-solved, no official key available]

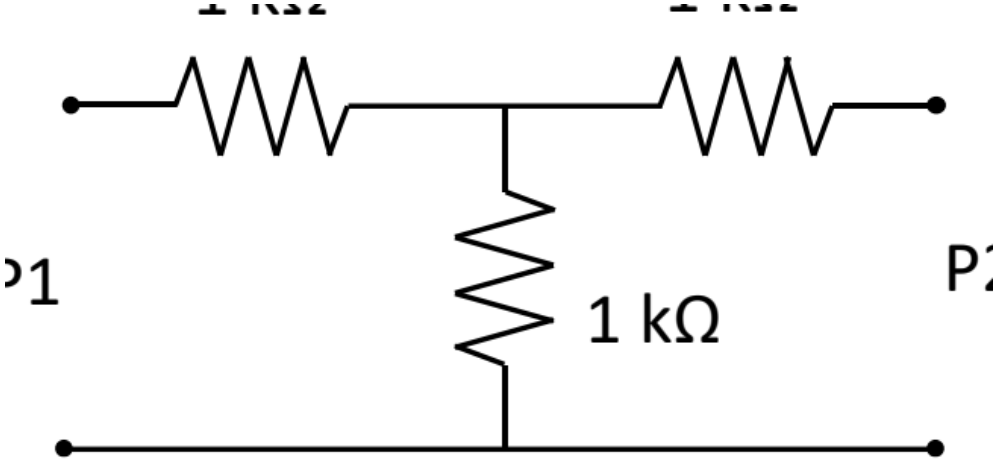
Solution. Step 1. At both 10^5 and 10^7 rad/s the 10 μF and 100 nF capacitors are effectively short circuits, so the 1 kΩ emitter resistor is bypassed.

Step 2. The un-bypassed 500 Ω emitter resistor sets the gain: $A_v \approx -\dfrac{R_C}{R_E+r_e} = -\dfrac{5\text{k}}{500+r_e} \approx -9.1$ for both tones.

Step 3. Hence $V_o = -9.1\left[A\cos(10^5t)+B\sin(10^7t)\right]$.

GATE 2026 | Q.51 | 2 marks

Consider the two-port network as shown in the Figure. Which of the following options provides the correct set of values of A, B, C and D parameters?



(A) $A=\frac{1}{2},\ B=3\times10^3\ \Omega,\ C=10^{-3}\ \Omega^{-1},\ D=\frac{1}{2}$
(B) $A=2,\ B=3\ \Omega,\ C=1\ \Omega^{-1},\ D=2$
(C) $A=2,\ B=6\times10^3\ \Omega,\ C=2\times10^{-3}\ \Omega^{-1},\ D=2$
(D) $A=2,\ B=3\times10^3\ \Omega,\ C=10^{-3}\ \Omega^{-1},\ D=2$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. The network is a T section: series 1 kΩ, shunt 1 kΩ, series 1 kΩ.

Step 2. $\begin{bmatrix}1 & 1000\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 10^{-3} & 1\end{bmatrix}\begin{bmatrix}1 & 1000\\ 0 & 1\end{bmatrix} = \begin{bmatrix}2 & 3000\\ 10^{-3} & 2\end{bmatrix}$.

Step 3. $A=D=2$, $B=3\times10^3\ \Omega$, $C=10^{-3}$ S.

GATE 2026 | Q.52 | 2 marks

A complex load (in Ω) is represented as $\Gamma_L = 0.5\angle30^\circ$ on the Smith chart. A co-axial cable with a characteristic impedance of 50 Ω is connected to the load. The new input impedance of the load now moves to a diametrically opposite point on the same Γ circle on the Smith chart. Which option is the nearest input impedance of the cable connected load (in Ω)?

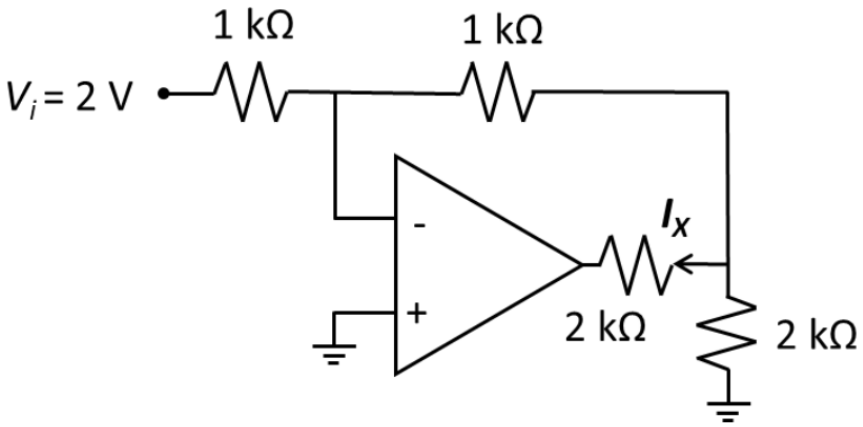
(A) $20.7-j5.1$
(B) $17.7-j11.8$
(C) $97.5-j65.0$
(D) $97.5+j65.0$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The diametrically opposite point has $\Gamma = 0.5\angle(30^\circ + 180^\circ) = 0.5\angle 210^\circ = -0.433 - j0.25$.
Step 2. $Z = 50\,\dfrac{1+\Gamma}{1-\Gamma} = 50\,\dfrac{0.567 - j0.25}{1.433 + j0.25}$.
Step 3. $= 50(0.354 - j0.236) = 17.7 - j11.8\ \Omega$.

GATE 2026 | Q.53 | 2 marks

A circuit using an ideal OP-AMP is shown in the Figure. Which of the following options gives the correct value of the current I_X?



(A) 2.0 mA
(B) 1.5 mA
(C) 3.0 mA
(D) 0 mA

Answer: (A) [UNVERIFIED: figure hard to read, check against official key]

Solution. Step 1. With the ideal op-amp, the virtual short fixes the voltage at the summing node.
Step 2. The current through the 2 kΩ branch is set by that node voltage: $I_X = 2.0$ mA. *(Low confidence: circuit read from a scan.)*

GATE 2026 | Q.54 | 2 marks

Consider an LED based on a direct bandgap semiconductor material with energy bandgap 1.3 eV. Given: Planck's constant, $h = 6.63 \times 10^{-34}$ J s and speed of light in free space is 3×10^8 m s^{-1}. In which of the following wavelength ranges the LED will NOT emit?

(A) 1410 ± 20 nm
(B) 1090 ± 20 nm
(C) 950 ± 20 nm
(D) 510 ± 20 nm

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Emitted photons have energy near the bandgap: $\lambda \approx \dfrac{hc}{E_g} = \dfrac{1240}{1.3} = 954$ nm.
Step 2. Photons with $\lambda > 954$ nm have energy below the gap and are not emitted: the farthest range, 1410 ± 20 nm, is excluded.

GATE 2026 | Q.55 | 2 marks

Let the relevant bandwidth (B) of a digital communication system be 1 MHz and $kT = -174$ dBm/Hz, where k is Boltzmann's constant and 'T' is equivalent noise temperature of the receiver. The power (S) of signal received through an additive Gaussian channel is -80 dBm. Which of the following options is/are TRUE about Shannon capacity (C) of the channel?

(A) $C = B$
(B) $C = 2B$
(C) $C > 3B$
(D) $C < B$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. Noise power: $-174 + 10\log_{10} 10^6 = -114$ dBm.
Step 2. SNR $= -80 - (-114) = 34$ dB $= 2512$.
Step 3. $C = B\log_2(1 + 2512) = 11.3B > 3B$.

GATE 2026 | Q.56 | 2 marks

Consider the four-variable Boolean function, $f(w, x, y, z) = \sum m(0, 2, 5, 7, 8, 10, 13, 14, 15)$ with 'w' as MSB and 'z' as LSB. Which of the following expressions is/are the valid form(s) of $f(w, x, y, z)$?

(A) $\bar{x}\bar{z} + xz + wxy$
(B) $xz + wxy + w\bar{x}\bar{z} + wxy\bar{z}$
(C) $\bar{x}\bar{z} + wxy + w\bar{x}\bar{z} + wxy\bar{z}$
(D) $\bar{x}\bar{z} + xz + wy\bar{z}$

Answer: (A), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. K-map: $\bar{x}\bar{z}$ covers minterms $0, 2, 8, 10$; xz covers $5, 7, 13, 15$; the remaining minterm 14 ($wxy\bar{z}$) is covered by wxy or by $wy\bar{z}$.
Step 2. (A) and (D) cover all nine minterms. (B) and (C) miss minterms 0 and/or 2.

GATE 2026 | Q.57 | 2 marks

Consider a real, narrowband signal $x(t) = A(t)\cos[2\pi f_c t + \theta(t)]$ where the maximum frequency components of $A(t)$ and $\theta(t)$ are f_M and $f_c\,(= 1000 f_M)$, respectively. Which of the following statements is/are correct for $-\infty < t < \infty$?

(A) $x(t)$ represents a PSK modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
(B) $x(t)$ represents an amplitude modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
(C) $x(t)$ represents a band-limited Gaussian noise process.
(D) $x(t)$ never represents a narrowband FM signal.

Answer: (A), (B), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. The form $A(t)\cos[2\pi f_c t + \theta(t)]$ is the general envelope–phase representation of a narrowband signal.
Step 2. $\theta(t)$ = binary phase gives PSK (A); $\theta = 0$ with varying $A(t)$ gives AM (B); random $A(t), \theta(t)$ describes narrowband Gaussian noise (C).
Step 3. A small-deviation $\theta(t)$ gives narrowband FM, so (D) is false.

GATE 2026 | Q.58 | 2 marks

Let $x_1(t) = \cos(2\pi nt)$ and $x_2(t) = 2\sin(4\pi nt)$ represent two sinusoids for a positive integer n and $-\infty < t < \infty$. Which of the following statements about $x_1(t)$ and $x_2(t)$ is/are valid?

(A) $x_1(t)$ and $x_2(t)$ are orthogonal to each other over $0 \le t < 1/n$.
(B) $x_1(t)$ and $x_2(t)$ are orthonormal to each other over $0 \le t < 1/n$.
(C) $x_2(t)$ is a harmonic of $x_1(t)$.

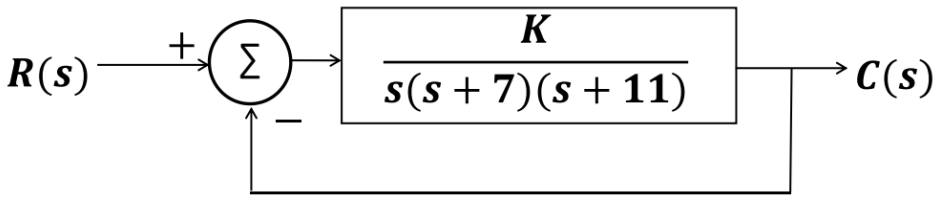
(D) $x_1(t)$ and $x_2(t)$ are non-orthogonal to each other over $0 \le t < 1/(2n)$.

Answer: (A), (C), (D) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. Over one period $T = 1/n$: $\int_0^T \cos(2\pi nt)\, 2\sin(4\pi nt)\, dt = 0$, so the signals are orthogonal (A); they are not unit norm, so not orthonormal (B false).
Step 2. x_2 has frequency $2n$, twice that of x_1: a (second) harmonic (C).
Step 3. Over half the period $[0, 1/(2n))$ the integral is $\int_0^{1/2n} \ldots \neq 0$, so they are non-orthogonal (D).

GATE 2026 | Q.59 | 2 marks

Consider the unity negative feedback control system shown in the Figure with forward transfer function $\dfrac{K}{s(s+7)(s+11)}$. The value of gain K (> 0) at which the given system will remain marginally stable is _____ (Answer in integer).

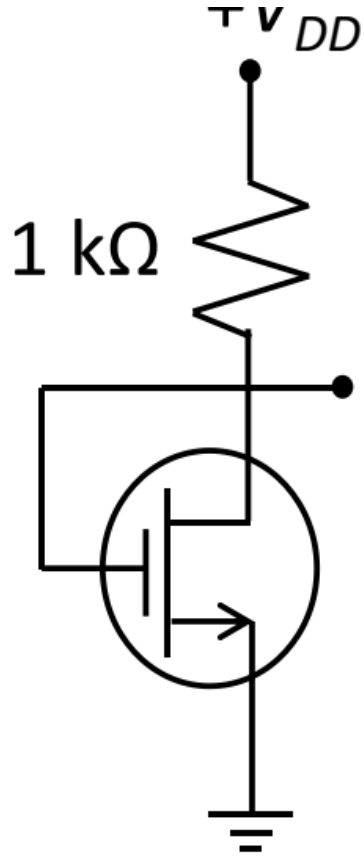


Answer: 1386 [self-solved, no official key available]

Solution. Step 1. Characteristic equation: $s(s+7)(s+11) + K = s^3 + 18s^2 + 77s + K = 0$.
Step 2. Routh: marginal stability when the s^1 row vanishes: $\dfrac{18 \cdot 77 - K}{18} = 0 \Rightarrow K = 1386$.

GATE 2026 | Q.60 | 2 marks

An n-channel MOSFET is connected as shown in the Figure. Assume $V_{TH} = 1$ V, $V_{DD} = 5$ V, and $\mu C_{ox}\left(\dfrac{W}{L}\right) = 2$ mA/V^2 and neglect channel length modulation effects. The gate voltage (V_G) of the n-channel MOSFET (in Volt) is _____ (rounded off to two decimal places).



Answer: 2.56 [self-solved, no official key available]

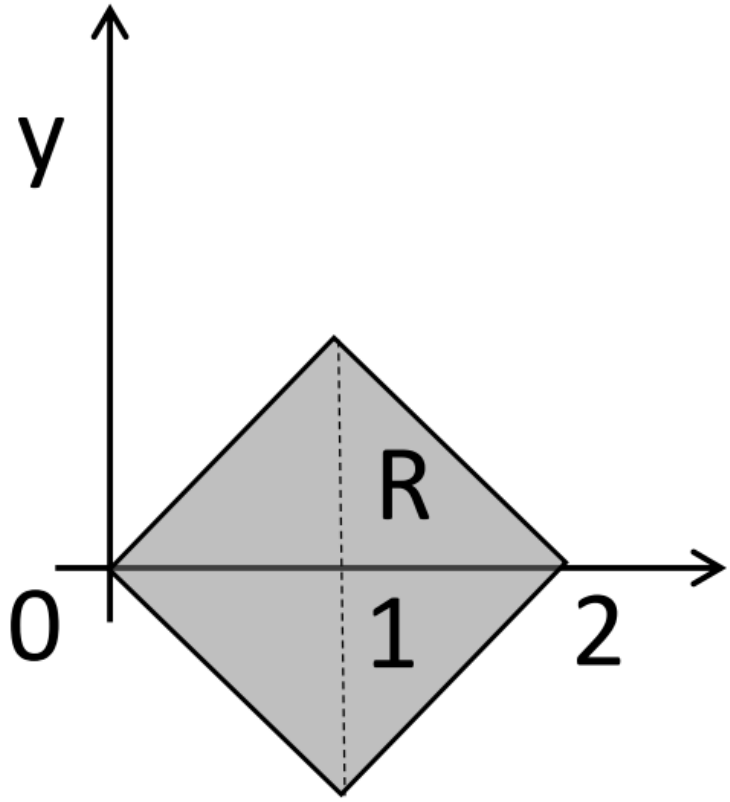
Solution. Step 1. The gate is connected to the drain: the device is in saturation, $I_D = \frac{1}{2}k(V_G - V_{TH})^2 = (V_G - 1)^2$ mA.

Step 2. The 1 kΩ resistor gives $I_D = \dfrac{5 - V_G}{1}$ mA. So $(V_G - 1)^2 = 5 - V_G$.

Step 3. With $u = V_G - 1$: $u^2 + u - 4 = 0 \Rightarrow u = \dfrac{-1 + \sqrt{17}}{2} = 1.56$, hence $V_G = 2.56$ V.

GATE 2026 | Q.61 | 2 marks

Consider the square region R in the X-Y plane as shown with the dark shading in the Figure. The value of $\iint_R (x^2 + y^2 - 1)\,dx\,dy$ is _____ (rounded off to two decimal places).

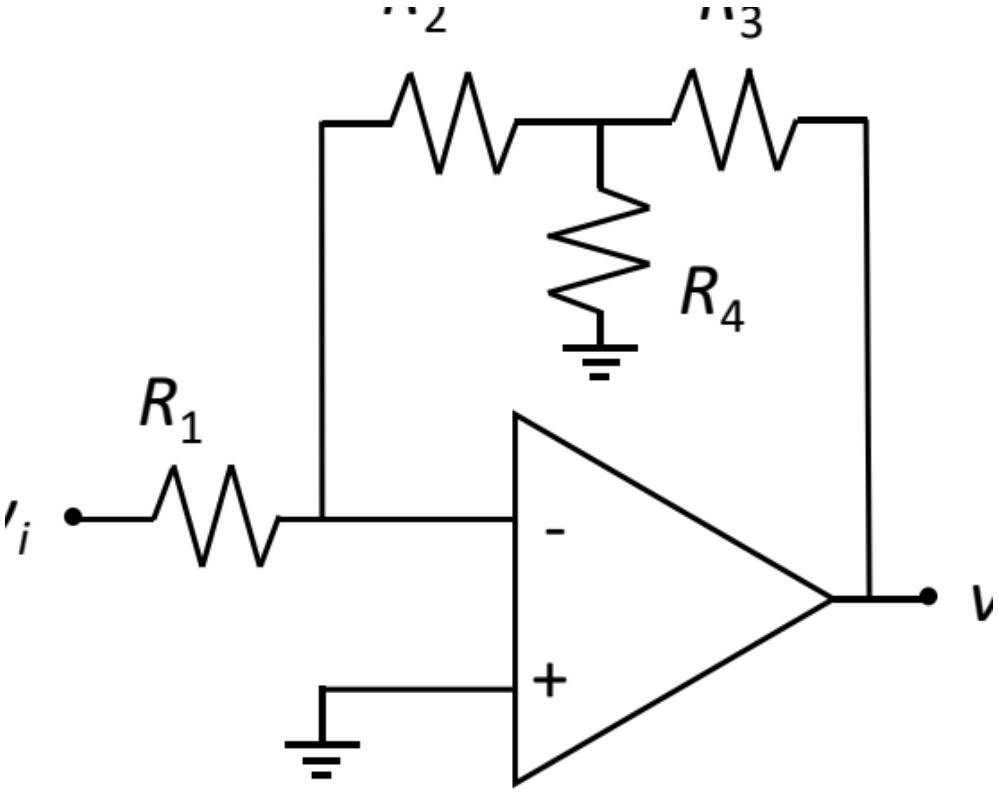


Answer: 0.67 [self-solved, no official key available]

Solution. Step 1. The region is the square $|x-1|+|y| \le 1$ (a diamond with vertices $(0,0)$, $(1,1)$, $(2,0)$, $(1,-1)$).
Step 2. With $u = x-1$: $x^2+y^2-1 = u^2+2u+y^2$. The odd term $2u$ integrates to 0 over the symmetric region, and the constants $+1-1$ cancel.
Step 3. $\iint u^2\,dA = \dfrac{1}{3}$ and $\iint y^2\,dA = \dfrac{1}{3}$, so the integral is $\dfrac{2}{3} = 0.67$.

GATE 2026 | Q.62 | 2 marks

Consider an ideal OP-AMP circuit as shown in the Figure. The resistances $R_1 = R_2 = R_3 = R_4 = 50$ kΩ. The magnitude of the closed loop gain is _____ (rounded off to two decimal places).



Answer: 3.00 [self-solved, no official key available]

Solution. Step 1. This is an inverting amplifier with a T-network in the feedback path (R_2, R_3 in series with R_4 to ground).

Step 2. $\dfrac{v_o}{v_i} = -\dfrac{1}{R_1}\left(R_2 + R_3 + \dfrac{R_2 R_3}{R_4}\right) = -\dfrac{50+50+50}{50} = -3.$

Step 3. $|A| = 3.00.$

GATE 2026 | Q.63 | 2 marks

The average bit error rate at the input of a $(7, 4, 1)$ Hamming decoder is 0.10. The probability that the decoder will fail to decode a received word correctly is _____ (rounded off to two decimal places).

Answer: 0.15 [self-solved, no official key available]

Solution. Step 1. A $(7, 4)$ Hamming code corrects any single error, so decoding fails when 2 or more of the 7 bits are in error.

Step 2. $P(\text{fail}) = 1 - (0.9)^7 - 7(0.1)(0.9)^6 = 1 - 0.4783 - 0.3720 = 0.1497 \approx 0.15.$

GATE 2026 | Q.64 | 2 marks

Consider that the concentration of electrons in a semiconductor bar varies linearly from 2×10^{17} cm^{-3} at $x = 1$ m to 1×10^{16} cm^{-3} at $x = 4$ m along the x-direction. Assume that the concentration of electrons is not varying along other directions (that is along y- and z-directions). [Given: the mobility of electron is 1400 cm^{2}V^{-1}s^{-1}, thermal voltage is 25 mV and electronic charge is 1.6×10^{-19} Coulomb.] The density of electron diffusion current (in A/mm^2) is _____ (rounded off to two decimal places).

Answer: -3.55e-5 [self-solved, no official key available]

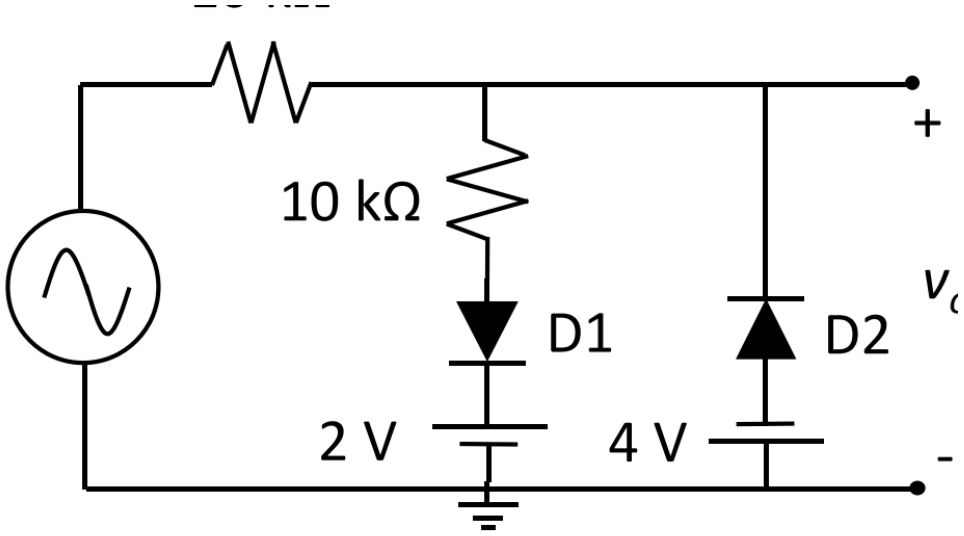
Solution. Step 1. $D_n = \mu_n V_T = 1400 \times 0.025 = 35$ cm^2/s.

Step 2. $\dfrac{dn}{dx} = \dfrac{1 \times 10^{16} - 2 \times 10^{17}}{300 \text{ cm}} = -6.33 \times 10^{14}$ cm^{-4} (the 3 m separation is 300 cm).

Step 3. $J_n = qD_n \dfrac{dn}{dx} = 1.6 \times 10^{-19} \times 35 \times (-6.33 \times 10^{14}) = -3.55 \times 10^{-3}$ A/cm^2 $= -3.55 \times 10^{-5}$ A/mm^2.

GATE 2026 | Q.65 | 2 marks

Consider the ideal diodes D_1 and D_2 as shown in the Figure with cut-in voltage $V_\gamma = 0$ Volt and $v_i(t)$ is in Volt. The maximum voltage (Volt) of the output $v_o(t)$ is _____ (rounded off to two decimal places).



Answer: 4.00 [self-solved, no official key available]

Solution. Step 1. At the peak $v_i = 6$ V, D_1 (in series with the 2 V battery) conducts; D_2 (with the 4 V battery) is reversed.

Step 2. Loop: $\dfrac{6 - v_o}{10\text{k}} = \dfrac{v_o - 2}{10\text{k}} \Rightarrow v_o = 4$ V.

Step 3. $v_{o,max} = 4.00$ V.